Poster: A Kinematic Root of Trust for Laser-Linked Satellites: Authentication via Delta-v Bounds over Time

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Abstract

The security of Optical Inter-Satellite Links (OISLs) rests on secret keys, a single point of failure. We ask what authentication anchor remains when every key leaks. The answer is the orbit itself. Reading, injecting, or modifying a laser crosslink requires physical presence inside a small attack zone. Every such seat lies off all natural orbits, so holding it burns fuel at a fixed rate. We price every attack in Δv, the velocity change a maneuver requires, paid in propellant from a fixed onboard supply. Assuming a monitored exclusion region around each terminal, our protocol tightens the attack zones and requires frequent active participation, turning this rate into a bill that grows over time. After a year, persistent impersonation costs 4 to 12 km/s of Δv, more than any spacecraft carries, while the defender spends a fraction of a percent of bandwidth. The physical Δv bound does not rely on computational hardness. Our preliminary results establish the feasibility of this approach under a single-attacker model and identify collusion resistance, operator authentication, and exclusion-zone enforcement as the main open challenges.

CCS Concepts

• **Security and privacy** → **Authentication**; **Security protocols**; **Trust frameworks**; • **Networks** → **Link-layer protocols**; *Physical links.*

Keywords

optical inter-satellite links; physical-layer authentication; orbital mechanics; delta-v; LEO constellations; position-based authentication

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1 Introduction

Optical Inter-Satellite Links (OISLs) rely on secret keys for authentication, but once those keys leak, no trusted secret remains to authenticate a replacement key. Compromised Commercial Off-The-Shelf (COTS) radios have leaked them outright [10], operational spacecraft run sensitive links either unencrypted or with weak cryptography [14], and quantum computing threatens the underlying mathematics. A laser link offers a way back to authentication that no radio link does. After a thousand kilometers, the beam is still only a couple of hundred meters wide, the receiver sees nothing outside a cone under a milliradian across, and both endpoints fly public, predictable orbits. Manufacturers market this geometry as inherent security, but only qualitatively [7], and surveys call Inter-Satellite Links (ISLs) an open Physical Layer Security (PLS) problem [1].

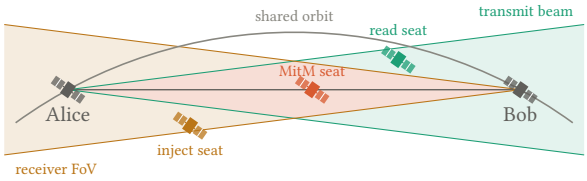


Figure 1: Every Optical Inter-Satellite Link (OISL) attack requires a physical seat: reading inside Alice's transmit beam, injecting inside Bob's receiver FoV, or rewriting from the on-axis MitM tube where the two overlap. Every MitM seat lies below the shared orbit, so an attacker must burn Δv without pause to hold it; a history-chained protocol turns that rate into a bill growing with time, pricing lasting impersonation above any spacecraft's fuel. Authentication then rests on orbital mechanics, not secret keys. (Angles exaggerated.)

The existing work leaves three gaps. (1) It focuses mainly on confidentiality. Keyless wiretap and exclusion-zone schemes bound what an eavesdropper can read [9], but a compromised satellite must first be authenticated. Any rekeying, and even Quantum Key Distribution (QKD), must bootstrap from an authenticated classical channel. (2) The existing physical-layer authentication schemes remain statistical. They validate Doppler or learned signatures with false-alarm and missed-detection rates and need trained reference data [13]. (3) Prior work does not quantify the attacker's physical effort. Δv, the velocity change a maneuver needs and the budget that binds every spacecraft, has never been tied to a security guarantee.

Our core insight is that every OISL attack requires a physical seat in space, and maintaining that seat incurs a Δv cost. Figure 1 summarizes this geometry. From it, we derive a deterministic kinematic-continuity guarantee that holds even when the attacker is not detected. A satellite that has lost every secret still owns one thing no attacker can copy for free, its predictable natural motion in orbit. We present four aspects of our work. **(1) A geometric threat and attacker model** that maps each Dolev–Yao capability on an OISL into a physical seat. **(2) A Δv cost model** that identifies where

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each attack is possible, which seat is cheapest, and the continuous maneuvering required to hold it. **(3) A position-, motion-, and time-derived authentication primitive** that grows stronger with the communication history. It forces any surviving single-spacecraft attacker into a detectable on-axis Man-in-the-Middle (MitM) position and supports recovery after total key leakage without relying on a surviving secret. It uses existing mechanisms as building blocks to constrain the attacker to this unfavorable position. **(4) Initial concepts for an actively enforced exclusion zone** supported by space-to-space situational awareness, together with example attack costs and stated limits.

2 Every Attack Needs a Seat

The system model consists of two satellites, Alice and Bob, on a shared 550 km orbit with standardized terminals [11, 12], and one attacker who knows all cryptographic keys but does not control either satellite, its flight system, or its fresh randomness. Every capability of the classic network attacker [5] then comes with a position (Figure 1). To read, the attacker needs a seat inside the transmit beam, a cone that stays only a few hundred meters wide along the whole link. The bound holds even if we grant the attacker an unlimited telescope and the best receiver ever demonstrated [4]. Just outside that cone, noise and the modulation itself limit what any measurement can extract. To inject or spoof, the attacker needs a seat inside the receiver's view cone. This second cone points in the opposite direction and has its origin at Bob. To rewrite traffic, the attacker must sit on the line of sight itself, capture the beam, and re-emit a copy. Outside all these zones, the attacker has no channel at all. Every modeling choice that follows favors the attacker, so every bound below is a floor.

3 Every Seat Burns Fuel

The seats are expensive because of one geometric fact, drawn in Figure 2. The line of sight between two satellites is a straight chord, but their shared orbit is a curved arc that bulges above it. The beam corridor, therefore, runs below the orbit, and any seat inside it hangs below the nearest natural orbit that would carry a satellite along for free. Near a terminal, the gap is small, about 36 meters at a seat 1 km out, and it widens toward mid-link.

That gap is what an attacker pays for. A seat on the line of sight co-rotates with Alice and Bob at their angular rate but sits at a lower radius, where a free-falling satellite would have to move faster to stay in orbit. Moving too slowly for its altitude, the seat would sink, and only continuous outward thrust holds it in place. In our circular co-orbital, continuous station-keeping model, the linearized dynamics put that thrust at $3n^2d$ for an offset d, with n the orbital rate [3], so the price grows with the offset. The attacker therefore favors a seat right next to Alice or Bob, held at near-zero relative speed. This is exactly what the exclusion zone denies. The zone is the volume around each terminal within which we assume Alice and Bob can detect an intruder. The further it pushes the attacker away from either terminal, the larger the forced offset, and with it the fuel rate.

On the 500 km reference link, that price starts at 11 m/s of Δv per day at the cheapest seats next to a terminal ($x = 1$ km) and climbs into hundreds of m/s per day toward mid-link, where the gap is widest. Close to a cone's apex, reading, injecting, and the on-axis MitM all share that rate. The cone is sub-milliradian, so the attacker can climb less than half a meter against the 36 m gap, and the forced offset d stays effectively the full gap. Three scenario changes raise the price further. (1) A longer link widens the gap everywhere, because the arc bends away from the chord faster near Alice and Bob. (2) A larger enforced keep-out zone pushes the attacker further off-axis, into strictly more expensive seats. (3) If Alice and Bob do not share an orbit, the attacker still pays everything above and must additionally track their changing relative position. A full numerical propagation confirms the analytic floor.

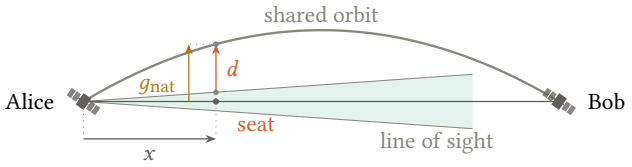


Figure 2: Why a seat costs fuel (not to scale). The shared orbit bulges above the straight line of sight, so a seat at distance x from a terminal sits a gap g_{nat} below it. Even after climbing to the top of the attack cone, the attacker maintains a forced offset d, and holding it requires continuous thrust.

4 A Guarantee That Grows With Time

A single position check can still be fooled for one moment. The protocol, therefore, stops treating authentication as a one-time event. After a first position-checked handshake, Alice and Bob re-authenticate frequently. Each round runs a fresh post-quantum key agreement (FIPS 203 [8]) for a key k_i and exchanges random bits, r_A from Alice and r_B from Bob, that only a receiver inside both beams can collect in full. Once a round has provided a secret, these bits travel encrypted under it. Everything folds into the next secret, $S_i = \text{KDF}(S_{i-1}, k_i, r_A, r_B)$, much like Signal's Double Ratchet. One missed round locks an impostor out for good because each round's key feeds every round that follows. Round by round, the positional bound tightens from the wide acquisition cone to the thin beam intersection and finally the line of sight, leaving the attacker only the hardest seat, the active on-axis MitM.

The round intentionally mixes two building blocks. The random bits are protected by geometry. Only a receiver inside both beams can see them, and no computation changes that. The encapsulated key is protected by mathematics and can be replaced only by an attacker actively sitting in the middle of both directions. An attacker must beat both, so the chain is as strong as the stronger gate. The physical Δv bound does not rely on computational hardness. Even if the Key Encapsulation Mechanism (KEM) is broken outright, the attacker still has to be there, inside both beams, every time. Each cone does brush the shared orbit at its wide end, beside and behind the terminal it points at, but those free seats see one beam only, so a free rider misses half of every round. For a persistent attacker, this turns the daily rate into an open-ended bill,

$$\Delta v_{\text{attacker}} \geq 3n^2 d\, T, \tag{1}$$

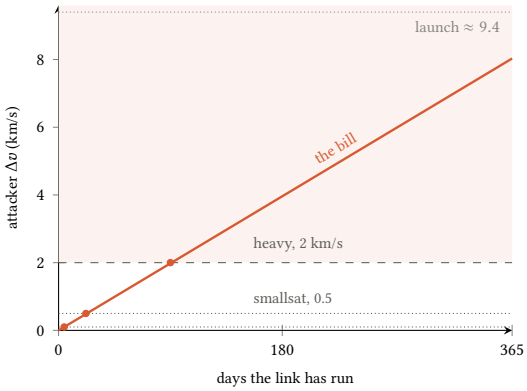


Figure 3: Impersonation is a bill that never stops. The cost of holding the seat increases linearly with time (representative 1,000 km link), while each attacker's fuel budget is fixed.

the radial station-keeping acceleration $3n^2d$ of the Clohessy–Wiltshire equations [3], held without pause and integrated over the age T of the link, with d the forced offset and n the orbital rate, 2π over the 96-minute period at 550 km. The same rounds also heal a key leak. Geometry, not an earlier secret, withholds the fresh bits from an absent attacker, so recovery needs no surviving secret. The link heals itself simply by continuing the protocol. Stated honestly, the guarantee is a disjunction. After time T, either the partner is genuine, or an attacker has paid the unbounded bill above and evaded surveillance throughout.

The protocol forces a surviving attacker onto the axis next to a terminal, at near-zero relative speed, short range, and a known direction. The link itself watches for the delay any relay adds to the round-trip time, for anomalies in its receive power, and for the glint any telescope throws back [6]. Enforcing the 1 km exclusion distance assumed above means watching only the thin cone per terminal, that holds every cheaper seat. This space-to-space situational awareness could run on hardware most satellites already carry, the star-tracker camera or the OISL terminal itself [2].

5 Preliminary Results

The attacker's Δv bill has no ceiling over time while every fuel tank is fixed, so time alone decides the contest (Figure 3). The underlying beam divergence is the diffraction-limited design value of the German Aerospace Center (DLR) CubeLCT terminal, flight-proven end-to-end on PIXL-1 and baselined for the CubeISL crosslink mission, so all costs rest on the smallest flight-proven laser terminal [11]. On a 1,000 km link, a chained seat costs about 22 m/s of Δv per day, so a 100 m/s CubeSat is spent in under a week, a 500 m/s smallsat in about three weeks, and even a 2 km/s heavy platform in about three months. A full year reaches 4 to 12 km/s across the link lengths we study, as much fuel as the launch to orbit, and no class of spacecraft can carry it today. The defender pays almost nothing, under 0.02% of the link bandwidth and a single small secret state. The shared orbit is the attacker's cheapest geometry, so these numbers are floors for the common Low Earth Orbit (LEO) link types. These bounds assume an attacker cannot remain within the monitored 1 km region around either legitimate terminal. With orbital mechanics and local proximity monitoring, a link can therefore regain authentication after hours to months, depending on Alice's and Bob's orbits.

6 Discussion and Future Work

Three limitations remain open and are left for future work: coordinated multi-spacecraft attacks; the primitive proving the spacecraft's position, not the operator behind it; and the exclusion zone resting on practice, not proof. Keeping the last cheap seats clear depends on space-to-space situational awareness sufficient to monitor a small volume of space beside each terminal. We believe this is the most consequential open problem on the path to a deployable guarantee. The refinement we most want to discuss is a joint budget showing that any attacker able to read or hold the link is also bright enough to be seen. Next steps could include wiretap coding paired with a linear program that balances the zone boundary against its bandwidth cost, a complete key-recovery protocol with stated guarantees, and an in-orbit demonstration on a CubeISL-class mission that also tests exclusion-zone detection.

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